

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
8	(a)	(i)	The energy of the photon becomes (1) PE and KE of electron (accept freeing electron for PE) (1) Alternative: Photon energy = KE + workfunction (1) Photon energy = KE + energy to release electron (2)	2			2		
		(ii)	Current / no. of electrons per sec / charge per sec etc. Accept power			1	1		
	(b)		Substituting correctly for r in equation 3 e.g. $p^2 = \frac{m_e e^2}{4\pi\epsilon_0 \frac{nh}{2\pi p}} (1)$ Correct algebra leading to $p = \frac{m_e e^2}{2\epsilon_0 nh} (1)$ (Minimum algebra: $p^2 = \frac{m_e e^2}{4\pi\epsilon_0 \frac{nh}{2\pi p}}$ QED i.e. cancelling π and p)		2		2	2	
	(c)	(i)	Minimum of 3 values substituted correctly (1) Answer = 13.55 [eV] or final substitution seen (1)	1	1		2	2	
		(ii)	Explain $n = 1$ e.g. leaving $n = 1$ or substituting $n = 1$ (1) Explain $n = \infty$ or PE at ∞ (1)		2		2		
	(d)		Valid method adopted e.g. using $n = 3$ and $n = 2$ to calculate the wavelength OR using 656 nm to confirm energy gap between $n = 2$ and 3 (1) $E = \frac{hc}{\lambda} = 3.03 \times 10^{-19} \text{ [J]} \text{ OR } 13.6 \times \left(\frac{1}{4} - \frac{1}{9}\right) \text{ OR } 1.89 (1)$ Final confirmation seen e.g. $3.03 \times 10^{-19} \text{ J} = 13.6 \times \left(\frac{1}{4} - \frac{1}{9}\right) \times 1.6 \times 10^{-19}$ OR $\frac{hc}{13.6 \times \left(\frac{1}{4} - \frac{1}{9}\right) \times 1.6 \times 10^{-19}} = 656 \text{ nm} (1)$			3	3	2	